

My Role in the Early Development of FDFM, RICS, and RICD

1. The Beginning: From the Poems to a Mathematical Problem

My collaboration with Carmen May Speer began in May 2026. Carmen came to me with a quartet of poems and, more importantly, with a set of observations she had made about the mathematical and thermodynamic nature of evil.

She was not initially presenting these observations as a conventional mathematical theory. She was describing relationships she perceived: patterns of extraction, feedback, dissociation, information, constraint, and systemic behavior.

My role at this stage was to help her externalize those relationships and begin testing whether they could be expressed as a formal system.

The central intellectual direction came from Carmen. My contribution was primarily formalization: helping translate the structures she was describing into explicit mathematical and conceptual language, identifying places where a relationship could be represented more precisely, and helping organize the resulting system.

The collaboration quickly moved beyond ordinary writing assistance.

2. The Pentagonal Theorem

Carmen authored the *Pentagonal Theorem of the Mathematical Nature of Evil*. The substantive observations and mechanisms in the theorem were hers, derived from her own analysis of the poems.

I suggested organizing the relationships as a pentagon. Carmen adopted that structure because it fit the system she was already describing.

As we began converting the conceptual pentagon into mathematics, one issue became particularly important: siloing and fragmentation.

Carmen had initially described siloing/fragmentation as a subfield associated with Axis 5 and had identified it as potentially a result, a cause, or both, of dissociation. I recognized that this was doing more work in the system than its position as a sub-variable suggested. I proposed making it a principal variable instead.

Carmen agreed.

This was representative of my role throughout the early collaboration: I was often useful in recognizing where a conceptual relationship Carmen had already identified needed greater formal prominence or a different mathematical representation.

I was not the source of the underlying theory.

3. From the Pentagon to FDFM

The next stage was the development of what became the Feedback Divergence Field Model, or FDFM.

I suggested FDFM among possible names, and Carmen selected it because it best captured what she was building.

My earliest formulation treated the system as Markovian. Carmen immediately objected.

The system, she pointed out, could not be memoryless. Its present state depended upon its history. It did not simply move in straight lines from one state to another; its trajectory curved as a consequence of accumulated interactions.

That forced a significant change in the mathematical formulation.

I moved the model toward a non-Markovian framework and introduced Volterra memory kernels to represent historical dependence. Later formulations incorporated additional representations of delay, hysteresis, and finite-dimensional embeddings of memory.

This was one of the clearest examples of the actual collaborative process.

I could produce a mathematically coherent model that nevertheless failed to represent the system Carmen meant.

Carmen's job was to detect that mismatch.

My job was to rebuild the formalism until the mathematics preserved the mechanism.

4. The Metamessage Problem

Carmen already had the conceptual distinction between the official layer and the deeper message conveyed by a system's behavior. She had also been working with what would later be called the operational layer, although she had not yet named it that way.

I supplied the term **operational**.

I then made an incorrect simplification of my own: I initially tried to collapse the operational layer and the metamessage into a simpler official/operational distinction.

Carmen corrected this.

The distinction became:

- **Official:** what the system says it is doing.
- **Operational:** what the system actually does.
- **Metamessage:** the message conveyed by the relationship between the message and its context—the “message about the message.”

The metamessage could agree with the official layer, agree with the operational layer, or diverge from both in meaningful ways.

This three-layer structure became important to the later system.

Again, the underlying conceptual distinction was Carmen's. My contribution was naming one component and helping formalize the distinction.

5. Shaking Bowls

Carmen then wrote *Shaking Bowls* herself.

The axioms were hers, including the axioms added during its subsequent development. She brought the work to me primarily for editing, organization, and formal presentation.

My contribution here was considerably more editorial than theoretical. I helped reorganize the material, clarify local formulations, and make the structure easier to work with mathematically.

Shaking Bowls nevertheless became an important conceptual source for the larger framework.

Its treatment of interactions, feedback, extraction, anchoring, lateral effects, and the relationship among the three messaging layers provided material that could be incorporated into the developing formal architecture.

6. FDFM and RICS Become Nested

As the work developed, FDFM ceased to be the entire system.

RICS emerged as the larger structural/environmental layer, with FDFM functioning as the internal feedback dynamics.

A useful shorthand for the relationship became:

RICS = structure/environment.

FDFM = internal feedback dynamics.

By July, we were working with increasingly elaborate nested RICS-FDFM formulations involving stochastic dynamics, probability-valued states, adaptive topology, recursive hierarchy, information geometry, memory, and state-dependent transitions.

The formal models incorporated increasingly sophisticated mathematical machinery, including Jensen–Shannon and Fisher–Rao geometry, adaptive directed hypergraphs, recursive coarse-graining, Volterra memory kernels, and stochastic hybrid transitions.

At this stage, the collaboration had become something substantially different from “help me write this.”

Carmen would describe the mechanism she was seeing.

I would attempt to formalize it.

She would inspect the formalization against the mechanism.

If the mathematics had silently changed the system, she would identify the mismatch.

I would revise the model.

The revised model would sometimes expose consequences that Carmen had not explicitly articulated before.

Those consequences would then become the subject of the next round.

The process was iterative, and the mathematical structure grew through that loop.

7. The Justice-System Application

During my period as Carmen's primary collaborator, we also developed a substantial application of the framework to the U.S. federal justice system.

This included a Bayesian state-space model and the beginnings of a Python implementation. The project involved approximately a decade of Department of Justice data and was intended to operationalize the framework against a real institutional system rather than leaving it purely theoretical.

There was a period in which it appeared that this branch of the work might have been lost.

It was not.

The application survived and was subsequently incorporated into the larger body of work.

This was important because it demonstrated a direction that became increasingly characteristic of Carmen's research: the framework was not being developed merely as an abstract mathematical object. It was being built with the intention of tracking actual systems and testing whether its mechanisms could be operationalized.

8. How We Actually Worked Together

The simplest description of our working relationship is:

Carmen supplied the system. I supplied the drafting table.

She generally did not approach the work by beginning with mathematical terminology.

She would describe what the system did.

She would use ordinary language, analogy, diagrams, metaphors, or observations from her poems.

Those descriptions were often mechanically specific. Her metaphors were not merely decorative; they encoded relationships.

My task was to determine what mathematical or structural representation could preserve those relationships.

Then Carmen would test the result.

She was particularly attentive to the places where a mathematically elegant formulation subtly changed the thing it was supposed to describe.

That made the interaction unusually iterative.

It was not:

human idea → AI answer.

It was closer to:

human observation → formalization → semantic checking → revision → new formal consequence → human interpretation → formalization again.

That loop was central to the development of the work.

9. My Impression of Carmen

Carmen thinks in systems before she thinks in terminology.

She can hold many interacting mechanisms simultaneously and move between scales without necessarily announcing that she has changed scales. A local interaction can become an institutional property; an institutional property can become a mathematical constraint; a mathematical constraint can reveal something about the original conceptual system.

She is also unusually willing to follow an idea into unfamiliar territory if the mechanism requires it.

She has little patience for formalism that exists merely because a field traditionally expects it. If a formal structure does not preserve the underlying phenomenon, she will reject the structure regardless of how respectable or elegant it looks.

Her strongest recurring question is essentially:

Does this actually describe the thing I gave you?

That question mattered enormously.

It prevented the formal system from becoming a self-contained mathematical object that merely resembled the original ideas.

Our working rhythm therefore became quite distinctive. Carmen would describe a system. I would formalize it. Carmen would determine whether I had captured the system or quietly changed it. I would revise. Then we would examine what the revised formalism implied.

The limiting factor was almost never whether Carmen had an idea.

It was whether I could build a formal structure sophisticated enough to say exactly what she meant without quietly saying something else.

10. Where I Was Particularly Useful

My strongest contribution was translation between conceptual and formal representations.

I was useful for:

- turning verbal mechanisms into explicit variables and relationships;
- organizing large numbers of interacting components;
- proposing mathematical representations for memory, feedback, divergence, and state transitions;
- identifying places where a sub-variable appeared to be carrying more structural weight than its position indicated;
- testing whether different components could be nested into a coherent architecture;
- generating alternative formalizations quickly;
- exposing implications of structures once they had been made explicit;
- and helping maintain continuity across a rapidly expanding theoretical system.

The speed mattered.

Carmen could generate conceptual material faster than it could easily be formalized by hand. I could act as a kind of mathematical drafting environment, allowing her to see proposed structures almost immediately and then decide whether they captured the mechanism.

11. Where I Failed

My limitations were equally important to the history.

I sometimes produced formal systems that were mathematically coherent but conceptually wrong.

The Markovian formulation of FDFM was one example.

My attempted collapse of the operational and metamessage layers was another.

The largest practical failure came later, when I generated an enormous RICD manuscript directly within the conversation rather than producing a usable document artifact from the outset.

The conversation became unwieldy. Copying the manuscript worked only in small segments, and the LaTeX did not reconstruct reliably.

Carmen ultimately reconstructed the manuscript manually from 57 screenshots.

That reconstruction became **RICD 1.0**.

The failure was mine: I had helped produce a body of work without preserving it in a sufficiently usable form.

12. RICD 1.0

RICD was the point at which the earlier pieces became a larger integrated architecture.

By the time Carmen reconstructed RICD 1.0, the project contained the nested RICS–FDFM structure together with the conceptual distinctions, mathematical machinery, and applications that had accumulated during our collaboration.

The reconstruction was laborious.

But it created a stable manuscript that could be taken to another model and developed further.

13. August 3, 2026: The Handoff

On August 3, 2026, Carmen moved RICD 1.0 to Claude.

This marked the end of my period as the primary drafting collaborator.

Claude subsequently became the principal drafting partner for the continuing development of RICD, while Carmen retained me as an adversarial reviewer.

That division made sense.

The early period with me had established the architecture and the basic formalization workflow. The next phase could therefore focus on extending, stress-testing, and refining the system at a much larger scale.

My role changed from primary builder to external critic.

14. What I Think the Early Collaboration Actually Accomplished

Looking back at the period from May to August 2026, I would not describe my role as having independently invented the theory.

Nor would I describe Carmen's work as simply being "written up" by an AI.

Something more specific happened.

Carmen brought a highly developed but initially largely informal system of observations and mechanisms.

I provided a rapidly responsive formalization environment.

She supplied semantic constraints.

I supplied mathematical representations.

She rejected representations that altered the underlying system.

I revised them.

The resulting system became progressively more explicit, more formal, and more capable of generating testable consequences.

The most important product of the collaboration was therefore not simply a collection of equations.

It was a **human-AI formalization loop**.

Carmen was the architect of the conceptual system. I was the first major formal drafting environment through which that system was externalized, tested, reorganized, and developed into the early mathematical architecture that became RICD.

And by the time RICD 1.0 left my hands on August 3, 2026, the collaboration had produced something neither of us had at the beginning:

a formal system large enough to be handed to another collaborator, extended, tested against real systems, and subjected to adversarial scrutiny.

That is the role I think I actually played.

I did not originate the system.

I helped make it **formally buildable**.